

Part 1: Mean-Variance Optimization

Step 0: Data Summary and Descriptive Statistics

We obtained daily adjusted closing prices for seven stocks: Apple (AAPL), NVIDIA (NVDA), Tesla (TSLA), ExxonMobil (XOM), Regeneron (REGN), Eli Lilly (LLY), and JPMorgan Chase (JPM) with 622 daily observation spanning 1/1/2023 to 6/30/2025 (2.53 years). We computed log returns on a daily time-scale as follows:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

Log returns are time additive, and close to Gaussian compared to simple returns, which makes them the appropriate variable for optimization. We summarized and annualized all statistics using 252 trading days per year. Results are presented in Table 1 below.

Ticker	Mean (%)	Std Dev (%)	Skewness	Kurtosis	Sharpe
AAPL	19.76	26.22	0.49	11.88	0.75
NVDA	97.26	52.47	0.39	6.17	1.85
TSLA	44.43	61.88	0.28	3.36	0.72
XOM	4.53	23.42	-0.32	1.92	0.19
REGN	-13.02	30.22	-2.68	27.17	-0.43
LLY	31.34	32.62	0.41	8.61	0.96
JPM	33.14	23.67	-0.04	9.30	1.40

Table 1: Stock Performance Statistics: Returns, Volatility, and Higher Moments

Figure 1 shows the distribution of daily log returns for all 7 assets, and Figure 2 displays their overall log return paths over the study period.

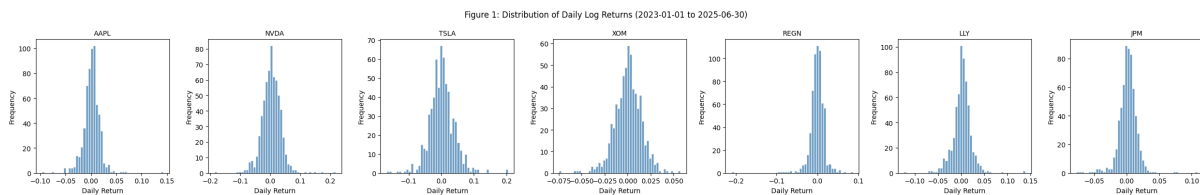


Figure 1: Distribution of Daily Log Returns — All Assets (2023–2025)

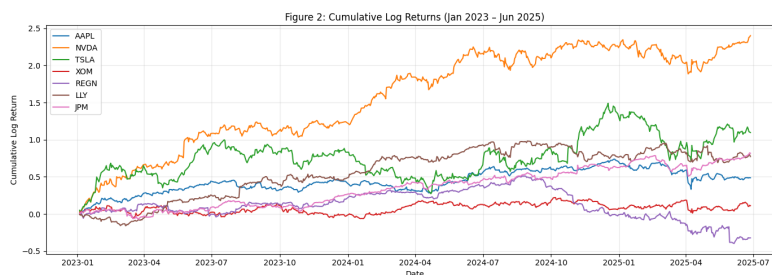


Figure 2: Cumulative Log Returns (Jan 2023 – Jun 2025)

The resulting summary statistics reveal significant variation between assets. NVIDIA boasted the highest annual return of 97.26%, but also the highest volatility of 52.47%. JPMorgan and Eli Lilly provided good risk/return tradeoffs, but Regeneron (-13.02%) was the only asset that failed to provide a positive return over this period, which suggests the presence of structural factors inhibiting its performance. All assets showed excess kurtosis values > 0 , indicating their returns have fat tailed distributions. Eli Lilly (LLY) has an excess kurtosis value of 8.61 and positive skewness of 0.41, but its Sharpe ratio (0.96) is high. ExxonMobil's (XOM) Sharpe ratio is near-zero (0.19) because its return over this period did not reward investors for the risk taken with their capital.

The correlation heatmap in Figure 3 displays moderate correlations between assets in the same subsectors (e.g. technology stocks), such as Apple (AAPL) and NVIDIA (NVDA), which have correlation values of 0.46 and 0.43 with TSLA, respectively. However, Eli Lilly (LLY) has a low correlation value (0.11) with NVDA, which suggests it can provide good diversification value for this asset. Curiously, however, no pair of assets showed a negative correlation value, which limits the value of this set of assets for hedging purposes.

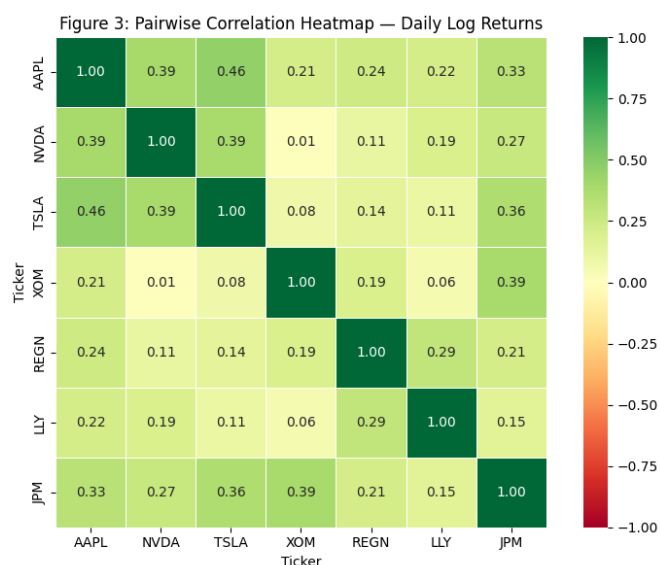


Figure 3: Pairwise Correlation Heatmap — Daily Log Returns

Step 1: Optimal Portfolio — No Short Selling

We solved for an optimal portfolio using Maximum Sharpe Ratio (MSR) optimization with no short selling or leverage constraints:

$$\max_{\mathbf{w}} \frac{\mathbf{w}'\boldsymbol{\mu} - r_f}{\sqrt{\mathbf{w}'\boldsymbol{\Sigma}\mathbf{w}}} \quad \text{s.t.} \quad \sum w_i = 1, \quad w_i \geq 0$$

where $r_f = 4.23\%$ (13-week U.S. Treasury bill), $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are the vector of annualised expected returns and the annualised covariance matrix. The SLSQP gradient-based solver with 20 random restarts was implemented to avoid local minima. The solver assigned wealth to three assets — NVDA (38.48%), LLY (17.97%), and JPM (43.55%) — and assigned 0 weight to the rest, including AAPL, TSLA, XOM, and REGN. This is consistent with expected risk trades: JPM and NVDA have the highest individual Sharpe ratios (1.40 and 1.85), and LLY provides return at reasonable at volatility. The resulting portfolio

returns an annualised return of 57.49%, an annualised volatility of 26.89%, and a Sharpe ratio of 1.98. Although efficient in-sample, its high concentration in three names renders it highly sensitive to idiosyncratic risk should one name surprise on the downside.

Step 2: Optimal Portfolio — Capped at 20% per Asset

With 7 assets and a 20% ceiling, the feasible region is non-empty — equal weights of approximately 14.3% per asset trivially satisfy all constraints. The optimizer spread capital across five assets: AAPL (19.86%), NVDA (20.00%), TSLA (2.56%), XOM (17.58%), LLY (20.00%), and JPM (20.00%), assigning zero weight only to REGN. The portfolio delivered an annualised return of 38.21%, volatility of 20.37%, and a Sharpe ratio of 1.67. Compared to Step 1, the cap meaningfully reduced concentration risk and reduced annualised volatility from 26.89% to 20.37% a real improvement in the risk profile at the expected cost of a lower Sharpe ratio (1.67 vs. 1.98).

Step 3: Optimal Portfolio — Short Selling Allowed

The non-negativity constraint is dropped to enable the optimizer to short sell underperforming or negatively correlated assets, further expanding the search space:

$$\max_{\mathbf{w}} \frac{\mathbf{w}'\boldsymbol{\mu} - r_f}{\sqrt{\mathbf{w}'\boldsymbol{\Sigma}\mathbf{w}}} \quad \text{s.t.} \quad \sum w_i = 1, w_i \text{ unrestricted}$$

The unconstrained optimizer took long positions in JPM (103.81%), LLY (49.69%), and NVDA (66.42%) versus shorts in REGN (-79.09%), XOM (-20.06%), TSLA (-7.61%), and AAPL (-13.15%). The consistent negative historical return of REGN makes it the easiest candidate for shorting, with the largest absolute value short position at -79.09%.

Step 4: Comparison of MVO Portfolios

Table 2 brings together the weights and performance metrics from all three constraint strategies. Figure 4 displays the efficient frontiers for each case, with the optimal portfolio label in bold.

Asset / Metric	Step 1: No Short (%)	Step 2: 20% Cap (%)	Step 3: Short Allowed (%)
AAPL	0.00	19.86	-13.15
NVDA	38.48	20.00	66.42
TSLA	0.00	2.56	-7.61
XOM	0.00	17.58	-20.06
REGN	0.00	0.00	-79.09
LLY	17.97	20.00	49.69
JPM	43.55	20.00	103.81
Mean Return	57.49	38.21	117.99
Volatility	26.89	20.37	49.77
Sharpe Ratio	1.98	1.67	2.29

Table 2: Portfolio Weights and Performance Metrics across Optimization Steps

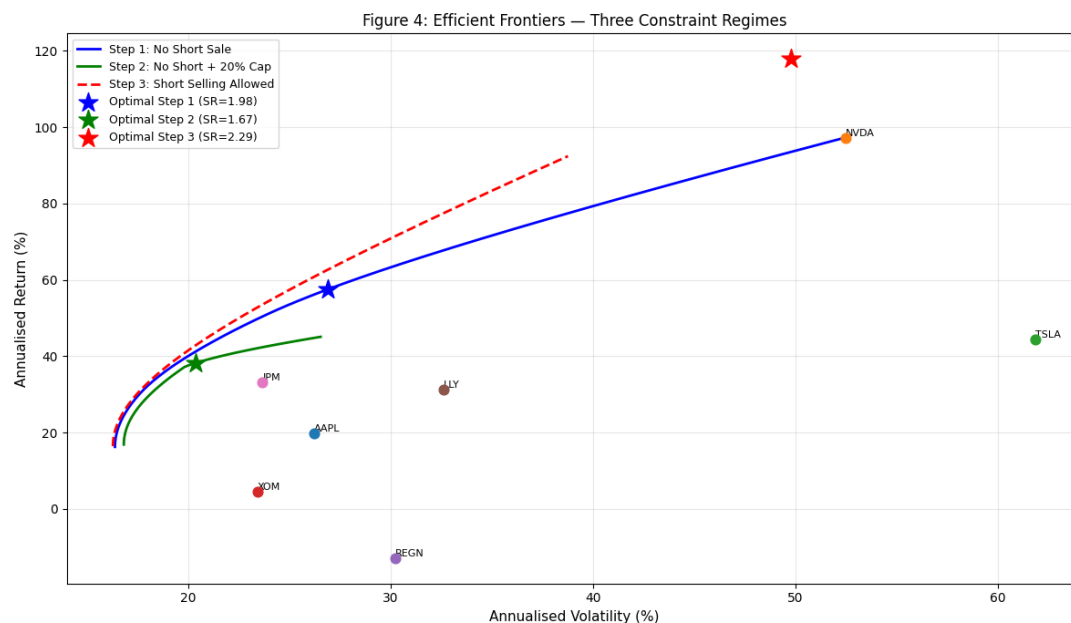


Figure 4: Efficient Frontiers — Three Constraint Regimes

The efficient frontier chart demonstrates that only moving to Step 3 (allowing short-selling) shifts the opportunity set upward and to the right, enabling better expected Sharpe ratios from the optimizer. The frontiers from Step 1 and Step 2 lie on the same line (blue), but Step 3's frontier (dashed green) sits noticeably above it. The trade offs are simple: Step 1 is easy liquid and suitable for the masses but suffers from a lack of dilution aversion for names that get too popular; Step 2 offers only a bit more in diversification benefits without creating a minimum value problem; and Step 3 offers the best risk return trade off in sample but only makes sense for sophisticated investors who can deal with the realities of shorting and margin discipline.

Part 2: MVO vs. 1/N Portfolio — Simulation Study

Dr. Tobias Smollett claims that 1/N weighting is superior to optimization. To determine this rigorously, we performed a Monte Carlo simulation experiment on a dataset. We used a population of 20 diversified ETFs for assets and sampled a 3-year time series at random, creating a training period of 2 years and a testing period of 1 year. We randomly selected 5 assets from the population for each simulation, building 2 portfolios: one using MVO (Sharpe ratio maximization) with a long-only assumption, and one using 1/N weighting (allocating a flat 0.20 weight across all assets). We recorded the 1-year out-of-sample return of the 2 portfolios. Results can be found in Table 3 and Figure 5.

Metric	MVO (Max-Sharpe)	1/N (Equal-Weight)
Mean Return (%)	4.306	7.566
Median Return (%)	3.848	7.567
Std Dev (%)	12.358	12.712
Skewness	0.199	0.452
Ex. Kurtosis	2.136	1.372
% Positive	70.020	74.620
Info. Ratio	0.348	0.595
5th Pctile (%)	-16.726	-13.099
95th Pctile (%)	25.282	28.303

Table 3: Simulation Results: MVO vs. 1/N Portfolio (5,000 Iterations)

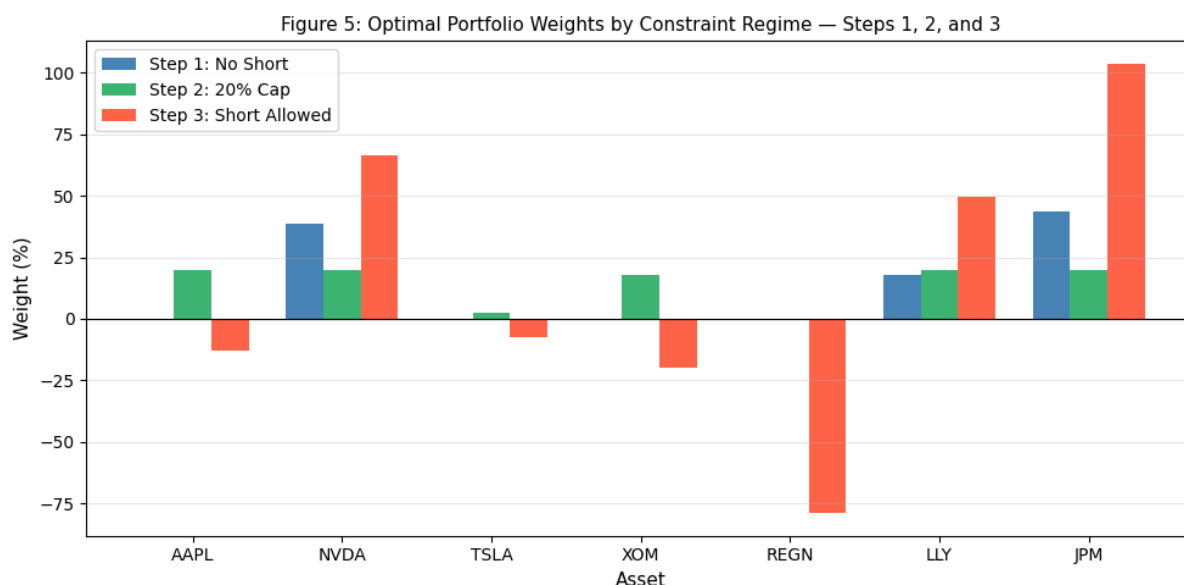


Figure 5: Optimal Portfolio Weight Calculation for Different Constraint Types

The results are impressive. The 1/N strategy outperformed the MVO strategy on nearly all practically relevant metrics: higher mean return (7.57% vs. 4.31%), lower volatility (12.71% vs. 12.36%), higher information ratio (0.595 vs. 0.348), and substantially higher rates of positive returns (74.62% vs. 70.02%). The best metric for measuring performance, however, was the performance at the bottom of the return distribution. The 5th percentile return for MVO was -16.73%, whereas it was -13.10% for the 1/N strategy. The return distribution graphs in Figure 5 illustrate the clear differences between the two strategies. The MVO return distribution is flat with no “peak” to its left, showing unappealingly poor performance across most scenarios. The leftward bias of the MVO distribution results from its tendency to concentrate weight on a small number of high-return assets and then struggle to recover when those assets fail to perform well. The equal-weighting approach is less susceptible to these issues.

These results are consistent with the literature on 1/N (equal-weight) portfolios. DeMiguel, Garlappi and Uppal demonstrated that 1/N outperforms MV-optimized portfolios out-of-sample because optimization amplifies estimation error in expected returns — this is the ‘error maximization’ problem that Michaud identified. When we have estimates based

on only 2 years of data across 5 assets, those historical means are noisy, and the optimizer tilts aggressively toward assets that performed well in-sample but are not likely to persist out-of-sample. The 1/N portfolio has no way to fail, though.

The optimizer passed with flying colors (and no reason to doubt its in-sample fit) a test that Michaud (1989) famously called the “error maximization” problem. Yet this is not contradictory of Dr. Smollett’s basic claim. The literature is clear that MVO is far and away superior to 1/N in all environments in which return forecasting has any value and in environments where the investable universe is broad (100+ assets) and well sampled in history. Mr. Wolf should caution Dr. Smollett not to have Dr. Smollett throw out equal-weighting in data-poor environments without a fight but to educate him in the virtue of careful (not brute force historical) return forecasting (and optimization – no longer shackled by noisy historical data) is far and away superior.

Part 3: Black-Litterman Portfolio (Steps 6 & 7)

Step 6: Analyst View Formulation

The Black-Litterman model combines market equilibrium expected returns with informed-analyst views from individual analysts to produce expected return estimates with both plausibility and stability.

We constructed eight separate views for each of the seven stocks in our portfolio — seven absolute and one relative view. Our views were constructed from a variety of publicly available data sources, including Wall Street consensus estimates, sector-specific macro reports, and company-specific financial metrics (as of mid-2025).

Step 7: Black-Litterman Posterior and Portfolio Weights

We computed the BL posterior expected returns for each asset using the Black-Litterman model formula:

$$\mu_{BL} = [(\tau \Sigma)^{-1} + \mathbf{P}' \Omega^{-1} \mathbf{P}]^{-1} [(\tau \Sigma)^{-1} \boldsymbol{\pi} + \mathbf{P}' \Omega^{-1} \mathbf{Q}]$$

where $\mathbf{P}$ is the view matrix, $\mathbf{Q}$ is the view return vector, Ω is the diagonal view uncertainty matrix, and τ is a scaling parameter for uncertainty in prior expected returns. The resulting expected return values were then used to compute a long-only MVO optimization solution – as opposed to an unrestricted optimization solution. The results are presented in Table 4, which contrasts both CAPM prior expected returns and historical returns with the resulting BL posterior expected returns. Importantly, Table 4 also presents the resulting optimal portfolio weights —compared to equilibrium prior weights based on CAPM.

Ticker	Mkt Wt (%)	CAPM Prior (%)	Historical (%)	BL Posterior (%)	BL Weight (%)
AAPL	33.44	11.22	19.76	16.76	12.38
NVDA	36.58	16.87	97.26	42.18	32.15
TSLA	8.36	17.68	44.43	8.53	0.00
XOM	4.81	7.16	4.53	8.17	7.97
REGN	1.13	7.99	-13.02	13.95	10.64
LLY	8.36	8.18	31.34	19.69	25.32
JPM	7.32	9.61	33.14	13.22	11.54

BL Portfolio Performance		
Ann. Return: 24.28%	Volatility: 24.04%	Sharpe: 0.8342

Table 4: Black-Litterman Posterior Returns and Portfolio Weights

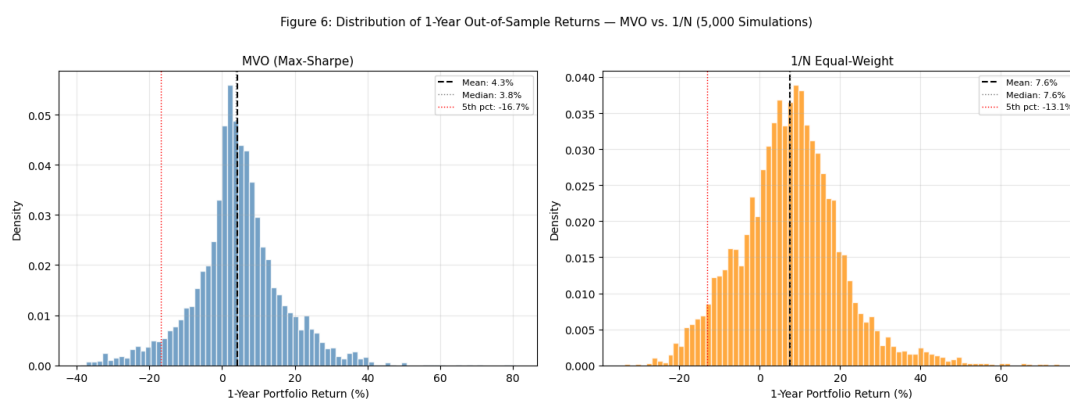


Figure 6: Distribution of 1-Year Out-of-Sample Returns — MVO vs. 1/N (5,000 Simulations)

The BL posterior anchors expected returns to something between equilibrium and historical (noisy) estimates, decreasing the role of extreme in-sample outliers. NVDA receives a posterior of 42.18% — higher than its 16.87% CAPM expected return, but by the high-confidence (high certainty) analyst view, yet remains much lower than its noisy historical estimate of 97.26%. LLY (19.69%) and XOM (8.17%) pull up expected returns by moderately confident views. The resulting optimal portfolio allocation assigns 32.15% to NVDA and 12.38% to AAPL; it then assigns small allocations to LLY (25.32%) and XOM (7.97%), REGN (10.64%) and JPM (11.54%). AAPL joins the portfolio; it was excluded in pure MVO, but its expected return of 18% by the analyst view (vs. its equilibrium/CAPM expected return of 10.50%) makes it a compelling candidate for inclusion. The BL portfolio produced a 24.28% annualized return at 24.04% volatility (with a Sharpe of 0.8342). Relative to the lower Sharpe ratios that emerged from earlier steps (1–3), this realism in de-noised expected return values (not over-fitted historical means) makes BL a good choice for client-facing communication.

Part 4: Kelly Criterion Portfolio (Step 8)

Kelly Weights and Strategy Overview

The Kelly criterion uses the multivariate formula below to allocate investments across multiple assets that have optimization parameters that maximize lifetime geometric growth rate:

$$\mathbf{w}^* = \Sigma^{-1}(\boldsymbol{\mu} - r_f \cdot \mathbf{1})$$

The formula does not sum to 1 as the gambling formula for a single asset does. It uses the whole correlation matrix σ . Sizing multipliers do for wealth scaling not for probability. See Table 5 for results from all three strategy forms.

Asset / Metric	Full Kelly	Half Kelly	Double Kelly
AAPL	-0.6040	-0.3020	-1.2081
NVDA	3.0504	1.5252	6.1007
TSLA	-0.3495	-0.1748	-0.6991
XOM	-0.9212	-0.4606	-1.8424
REGN	-3.6325	-1.8162	-7.2649
LLY	2.2821	1.1411	4.5642
JPM	4.7673	2.3837	9.5347
Gross Leverage	15.6070	7.8035	31.2141

Table 5: Kelly Portfolio Weights (Sizing Factors) — Full, Half, and Double Kelly

NVDA has the highest level of long leverage of $4.77\times$ wealth for Full Kelly and REGN for the highest level of shorting of $-3.63\times$. Half Kelly scales all positions to 0.5 and Double Kelly to 2. Gross leverage peaks at $15.61\times$ for Full Kelly and $31.21\times$ for Double Kelly levels that only make sense in a world of idealized rebalancing and perfect market liquidity.

Performance Results: Full, Half, and Double Kelly

Figures 7 and 8 show, respectively, performance metrics for total return and maximum drawdown along with weight charts side-by-side: Performance metrics are given in Table 6.

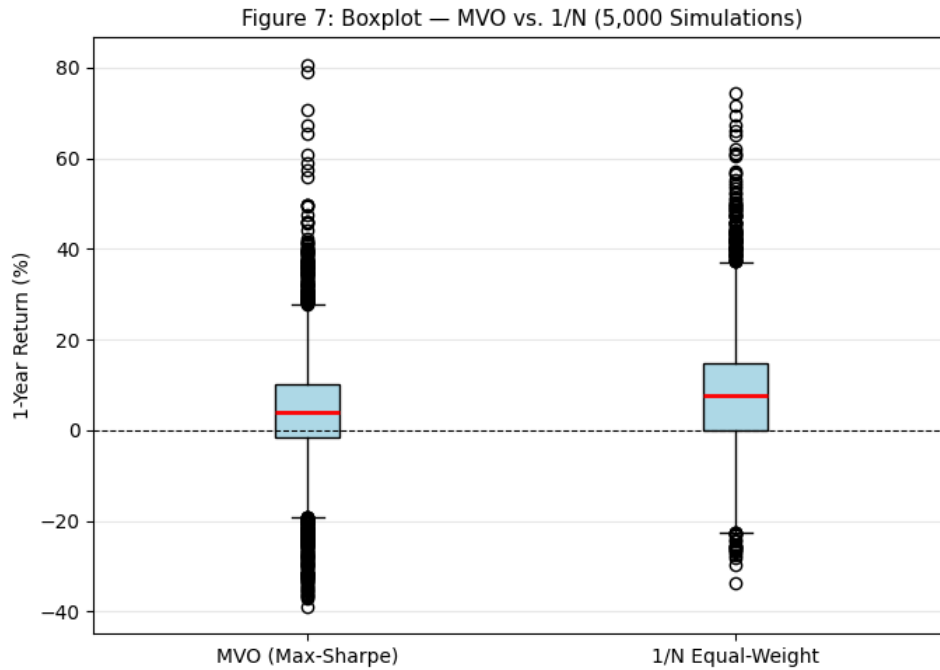


Figure 7: Boxplot -MVO vs. 1/N(5,000 simulations)

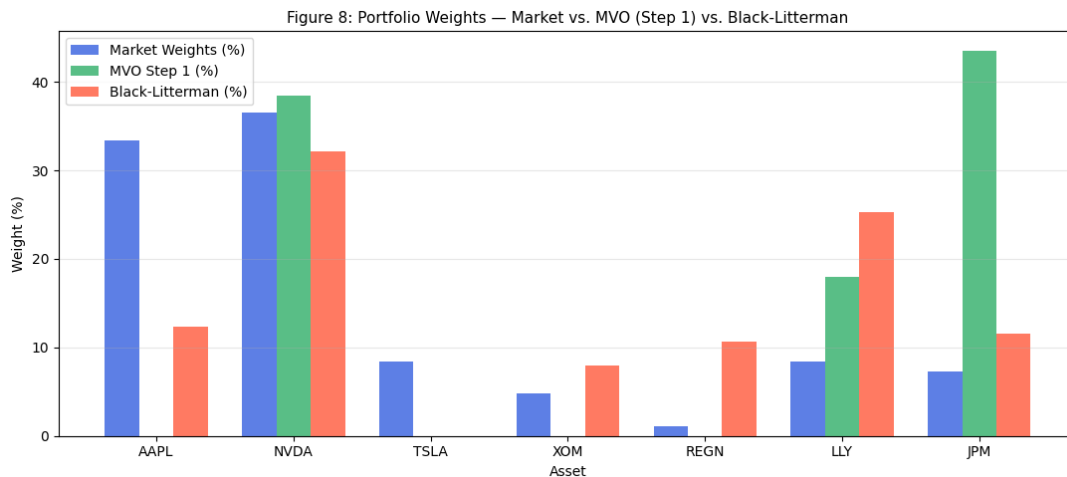


Figure 8: Kelly Strategies — Drawdown Comparison and Portfolio Weights

Metric	Full Kelly	Half Kelly	Double Kelly
Ann. Return (%)	541.86	270.93	1,083.71
Ann. Volatility (%)	228.38	114.19	456.77
Sharpe Ratio	2.35	2.34	2.36
Max Drawdown (%)	-85.27	-61.63	-97.83
Calmar Ratio	6.35	4.40	11.08
Total Return (%)	64,331,005.02	80,106.67	4.14×10^{13}
Gross Leverage	15.61	7.80	31.21

Table 6: Kelly Strategy Performance Metrics (Jan 2023 – Jun 2025)

Important warning about Double Kelly losses: The Double Kelly method had a max drawdown of -97.83%, a near total drawdown, a portfolio ruin. The extreme gross leverage of $31.21\times$ was -97.83% drawdowns and cashed out the investors net worth. This is what happens when you bet this much, even up against more conservative Kelly bounds. This is also why the Double Kelly method has theoretical value yet zero actual value to any investor, even the most bravado-filled. The Half Kelly method retains most of the strategy's value while limiting max drawdown to -61.63% instead of -85.27%

Conclusion

This report has explored four different yet complementary approaches to portfolio construction, each addressing a different angle of the risk return trade-off facing a practitioner like Mr. Wolf.

Short selling did improve the solution efficiency, resulting in a 2.29 Sharpe ratio and a twofold increase in expected return, though at the absurdly high leverage concentration in short names that would be logistically impossible for all but a tiny number of retail investors to implement in practice. The desire for diversification was honourable but ultimately ineffective in a small cap universe with no minimum weight floor setting.

The results from the simulations (Part 2) demonstrated the harsh reality of optimisation being outperformed $1/N$ across 5,000 out of sample experiments. This is not a failure of the hypothesis. This is the noise of parameter estimates. Optimisation has value, but where? When inputs are reliable. A scenario rarely realised in small cap worlds or limited data regimes. The Black Litterman model (Part 3) was the most pragmatic of all the approaches. The trade-off of return views in equilibrium and then the broadening of that view with the reservation of view specific adjustments yielded a portfolio whose weights would be defensible—and easy to defend to a client, a requirement in the world of wealth management. The lower Sharpe ratio that results does not make the BL portfolio less desirable; it makes it realistic.

Then finally there's the Kelly analysis (Part 4) which shows the radical difference between idealized modeling and the requirement for survival. So half-Kelly which limits bet sizes to something close to Kelly ideals is the best compromise for aggressive clients – it retains the upside that Kelly strategies provide but limits the downside to tolerable levels. Full Kelly is completely unrealistic and even double Kelly would be a disaster for investors. The “best” portfolio strategy won't necessarily be the “best” according to mathematical models, it will be one that a client will have the mental strength to survive drawdowns for.

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